

DIHWorld Audio

Harmonics / Pitch Technical Manual

Current product versions, controls, presets, meters, and placement

DIHWorld Audio builds connected production systems around the Imhotep 369 Engine—a Triadic Harmonic Lattice—and Ma'at 357 / MomentumBend. Phi-derived control and the canonical Phi/2 ceiling preserve movement, balance, and musical intent without brickwall limiting.

This edition documents 3 current products. Release numbers are taken from the installed DIHWorld Audio Audio Units verified on 2026-10-05; they are product versions, not a shared catalog version.

Canon reference: normalized $\Phi/2 = 0.809016994$. MomentumBend shapes motion and containment; the ceiling is a canonical boundary, not permission for hard limiting.

Operating law: begin from reset or a purpose-matched preset, compare at equal loudness, change one system at a time, and judge the result in the full arrangement.

Thoth

RELEASE	FAMILY	PLACEMENT
v1.0.1	Harmonics / Pitch	Insert Safe

Sacred harmonic EQ with low and high shelves, visible shelf curves, Flower of Life mode, protected I/O, and parallel mix.

Recommended use: harmonics/pitch

Customer formats: AU, VST3, Standalone

Exposed parameter surface

Parameter ID	Display name	Type
power	Power	Bool
input	Input	Float
mix	Mix	Float
ls_gain	Low Shelf	Float
hs_gain	High Shelf	Float
output	Output	Float
flower	Flower of Life	Bool
channelmode	Channel Mode	Choice
dualmonolink	Dual Mono Link	Choice
gainmatch	Gain Match	Bool

Factory starting points — source-derived purpose

THOTH FLAT REFERENCE

THOTH FLAT REFERENCE — Returns the harmonic field to a flat reference for hearing the source before a spectral move.

THOTH MIX POLISH

THOTH MIX POLISH — Sets THOTH's low/high harmonic lanes and Flower field for the named spectral job.

THOTH VOCAL CLARITY

THOTH VOCAL CLARITY — Frames vocal clarity/warmth/forwardness according to the named recipe while preserving the source center.

THOTH VOCAL WARMTH

THOTH VOCAL WARMTH — Frames vocal clarity/warmth/forwardness according to the named recipe while preserving the source center.

THOTH DRUM HARMONIC LIFT

THOTH DRUM HARMONIC LIFT — Uses the harmonic lanes to add drum lift, kick edge, or snare weight without a broad level boost.

THOTH BASS TRANSLATE

THOTH BASS TRANSLATE — Adds a harmonic voice to bass so it translates above the fundamental.

THOTH LOW MID CLEANUP

THOTH LOW MID CLEANUP — Cleans the low-mid lane before adding presence or air.

THOTH PRESENCE SEAT

THOTH PRESENCE SEAT — Seats upper articulation and air with a deliberate high-lane move.

THOTH GENTLE AIR

THOTH GENTLE AIR — Seats upper articulation and air with a deliberate high-lane move.

SOLFEGGIO 396 GROUND

SOLFEGGIO 396 GROUND — Uses the 396 Hz Solfeggio node as a grounded harmonic starting point; compare the actual level, not the number.

SOLFEGGIO 417 WEIGHT

SOLFEGGIO 417 WEIGHT — Uses the 417 Hz node as a weight/change-oriented harmonic starting point.

SOLFEGGIO 528 BODY

SOLFEGGIO 528 BODY — Uses the 528 Hz node as a body-oriented harmonic starting point.

SOLFEGGIO 639 SCRIBE

SOLFEGGIO 639 SCRIBE — Uses the 639 Hz node as a connection/vocal-support harmonic starting point.

SOLFEGGIO 741 FOCUS

SOLFEGGIO 741 FOCUS — Uses the 741 Hz node as a focus/clarity harmonic starting point.

SOLFEGGIO 852 AIR

SOLFEGGIO 852 AIR — Seats upper articulation and air with a deliberate high-lane move.

SOLFEGGIO 963 FLOWER

SOLFEGGIO 963 FLOWER — Uses the upper Solfeggio node as an air/open-field harmonic starting point.

THOTH DYNAMIC NOTCH SHOWCASE

THOTH DYNAMIC NOTCH SHOWCASE — Demonstrates a dynamic spectral contour so the harmonic movement can be heard in context.

THOTH DYNAMIC SMILE SHOWCASE

THOTH DYNAMIC SMILE SHOWCASE — Demonstrates a dynamic spectral contour so the harmonic movement can be heard in context.

THOTH DYNAMIC AIR SHOWCASE

THOTH DYNAMIC AIR SHOWCASE — Seats upper articulation and air with a deliberate high-lane move.

Meters and visualizers

Spectrum bands: Live energy in each displayed EQ region. Use to see where harmonic energy is accumulating.

Shelf curves: Current low/high shelf response. These show the selected EQ shape, not measured level.

IN / OUT: Pre/post peak activity. Use for level-fair harmonic decisions.

TEHUTI

RELEASE	FAMILY	PLACEMENT
v1.0.1	Harmonics / Pitch	Insert Safe

Harmonic motion processor using low, mid, and high ducking with depth, speed, mix, and Flower of Life mode. A named optional Sidechain provides external spectral-key detection; without a routed key, detection follows the main input.

Recommended use: harmonics/pitch

Customer formats: AU, VST3

Exposed parameter surface

Parameter ID	Display name	Type
power	Power	Bool
low_duck	Low Duck	Float
mid_duck	Mid Duck	Float
high_duck	High Duck	Float
depth	Depth	Float
speed	Speed	Float
mix	Mix	Float
flower	Flower of Life	Bool
gainmatch	Gain Match	Bool
channelmode	Channel Mode	Choice
dualmonolink	Dual Mono Link	Choice

Factory starting points — source-derived purpose

TEHUTI OPEN REFERENCE

TEHUTI OPEN REFERENCE — Returns all duck lanes to an open reference so sidechain behavior can be judged from scratch.

TEHUTI LOW SPEED HIGH DEPTH

TEHUTI LOW SPEED HIGH DEPTH — Deep, slow low-band movement for sustained bass/ducking; use it on a return or bus, not a transient source.

TEHUTI LOW DEPTH HIGH SPEED

TEHUTI LOW DEPTH HIGH SPEED — Faster, shallower low-band motion for rhythmic pocket without a long swell.

TEHUTI LOW BAND LOCK

TEHUTI LOW BAND LOCK — Locks the low band while leaving mid/high space mostly untouched.

TEHUTI MID SWIM FAST

TEHUTI MID SWIM FAST — Frames the mid band for vocal/chord separation; choose fast or slow according to the phrase rhythm.

TEHUTI HIGH SHIMMER DUCK

TEHUTI HIGH SHIMMER DUCK — Uses the high-band duck lane to protect air, hats, or shimmer without carving the whole mix.

TEHUTI FULL BAND PUMP

TEHUTI FULL BAND PUMP — Engages coordinated low/mid/high ducking for an audible full-spectrum pump.

TEHUTI REVERSE POCKET

TEHUTI REVERSE POCKET — Reverses the expected pocket so the movement blooms into the next hit.

TEHUTI VOCAL CARVE DEEP

TEHUTI VOCAL CARVE DEEP — Makes space for a vocal in the named depth while keeping the musical bands intact.

TEHUTI PERCUSSIVE SNAP

TEHUTI PERCUSSIVE SNAP — Uses faster envelope movement to clear percussion and preserve its attack.

TEHUTI SLOW WIDE BREATH

TEHUTI SLOW WIDE BREATH — Uses slower Depth/Speed for a breathing, organic return instead of hard pumping.

TEHUTI PARALLEL EDGE

TEHUTI PARALLEL EDGE — Places the movement on a parallel path so the dry signal retains its edge.

Meters and visualizers

INPUT: Main-program peak activity. Shows the signal being ducked.

SIDECHAIN SEND: External sidechain envelope. Confirms that the host sidechain is present.

LOW GR: Reduction below 162 Hz. Shows low-band ducking driven by the sidechain.

MID GR: Reduction from 162 Hz to 1.62 kHz. Shows mid-band ducking driven by the sidechain.

HIGH GR: Reduction above 1.62 kHz. Shows high-band ducking driven by the sidechain.

Duck curves: Expected reduction versus sidechain energy. The live dot shows the current operating point.

History: Recent sidechain and reduction motion. Use to judge pumping and release behavior.

AU

RELEASE	FAMILY	PLACEMENT
v1.0.1	Harmonics / Pitch	Insert Safe

DIHWorld instrument and synth with eight verified embedded factory sounds, optional user-source drag-and-drop, root detection, modulation, space, glide, sustain, and master control. Afro Horn Lift uses the embedded Warm Brass source for playable brass leads, lifts, and response lines. No Santuri source is claimed or exposed.

Recommended use: instrument/harmonic utility

Customer formats: AU, VST3

Exposed parameter surface

Parameter ID	Display name	Type
SOUND	Factory Sound	Choice
channelmode	Channel Mode	Choice
dualmonolink	Dual Mono Link	Choice
PTAH	PTAH	Referenced
RA	RA	Referenced
SET	SET	Referenced
ASAR	ASAR	Referenced
MAAT	MAAT	Referenced
SEKHMET	SEKHMET	Referenced
KHNUM	KHNUM	Referenced
HERU	HERU	Referenced
DRIVE	DRIVE	Referenced
AMOUNT	AMOUNT	Referenced
DEPTH	DEPTH	Referenced
SV_MIX	SV_MIX	Referenced
HARP	HARP	Referenced
ORGAN	ORGAN	Referenced
CHOIR	CHOIR	Referenced
MOD	MOD	Referenced
BREATH	BREATH	Referenced
VIBRATO	VIBRATO	Referenced
PHI	PHI	Referenced
AZIMUTH	AZIMUTH	Referenced
FLOWER	FLOWER	Referenced
HATHOR	HATHOR	Referenced
RAW	RAW	Referenced
MOD_MACRO	MOD_MACRO	Referenced
MASTER	MASTER	Referenced
GLIDE	GLIDE	Referenced
PURE	PURE	Referenced

Parameter ID	Display name	Type
SUSTAIN	SUSTAIN	Referenced
GAINMATCH	GAINMATCH	Referenced

Factory starting points — source-derived purpose

AU Core

AU Core — Returns the compact AU voice bank to its canonical starting sound.

Canon Glass

Canon Glass — Selects a clear, bright harmonic voice for articulate keys or plucks.

Temple Keys

Temple Keys — Selects a keyboard/organ foundation with the harmonic lanes kept stable.

Organ Foundation

Organ Foundation — Selects a keyboard/organ foundation with the harmonic lanes kept stable.

Motion Pluck

Motion Pluck — Selects a short, focused pluck voice with Drive available for edge.

Horizon Pad

Horizon Pad — Selects an open sustained pad voice for a wider harmonic field.

Root Foundation

Root Foundation — Selects a centered root voice for building a low or tonal foundation.

Afro Horn Lift

Afro Horn Lift — Selects a forward horn-like harmonic voice for rhythmic or Afro-inspired lifts.